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Original Research Article (Experimental)

Clinical consensus statements on *Marsha Nasya* – A feasibility study towards developing clinical practice guidelines of therapeutic procedures in Ayurveda

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ABSTRACT

Background: Modern pedagogical frameworks and recent advancements—such as quality accreditation and insurance coverage in Ayurveda—underscore the need for clear and practical guidelines for therapeutic procedures, particularly *Panchakarma*. The therapeutic procedures outlined in Ayurveda, notably *Panchakarma*, comprising five internal cleansing therapies (emesis, purgation, decoction enema, unctuous enema, and errhine therapy) currently lack published contemporary practice guidelines. Developing evidence-based, expert-endorsed clinical practice guidelines for these therapies would enhance their safety and effectiveness, ensuring their wider acceptance.

Ayurveda classical texts describe *Marsha Nasya* (MN), a therapy in which medicines mainly having an oil base are instilled in the nose in a comparatively higher dose, as a *Panchakarma* treatment that targets a range of pathological conditions, particularly those affecting the head and neck area.

Objective: This study aimed to develop a comprehensive set of evidence-based, expert-endorsed Clinical Consensus Statements (CCSs) for MN therapy utilizing the RAND/UCLA Appropriateness Method (RAM).

Methods: Best Practice Statements (BPSs) were developed based on an extensive literature review, inputs from physicians, and a survey conducted across multiple healthcare facilities. Nine experts participated in rounds 1 and 2 of RAM. The panelists rated each of these statements, and the statements that were deemed appropriate statistically were included in the final set of recommendations. The study was prospectively registered in the Clinical Trial Registry of India (CTRI number: CTRI/2023/05/052422 dated May 09, 2023) and PREPARE (Practice Guideline Registration for Transparency), an international prospective register for clinical practice guidelines. (PREPARE-2024CN612).

Results: A consensus was reached on 63 BPSs across six domains: the general domain, the preparatory procedures domain, the main therapeutic procedure domain, the post-therapy procedures domain, the dosage domain, and the quality assurance domain.

Conclusions: The detailed, expert - endorsed and evidence - based statements represent the important procedural aspects of MN and can be the basis for further clinical validation and practice of MN. The study may serve as a model for developing evidence-informed clinical practice guidelines for *Panchakarma* therapies and other therapeutic procedures in Ayurveda and can offer valuable guidance to clinicians, researchers, students, and policymakers.

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1. Introduction

Systematised practising methods and quality assurance are prerequisites for an effective therapeutic procedure. Ayurveda, the traditional medical system in India, employs an array of therapeutic procedures for managing various diseases, of which *Panchakarma* is considered vital. Ayurveda texts advocate patient selection and the administration of therapies in *Panchakarma* (five internal bio-cleansing therapies), which are to be carried out with extreme caution and under the strict supervision of qualified physicians [1]. The classical texts of Ayurveda describe these therapies in various contexts; however, abstract descriptions, which are in the form of Sanskrit verses, are not often interpreted adequately and practised inconsistently across diverse healthcare settings.

Efforts towards standardising these procedures have been undertaken at various levels [2]. However, such initiatives are rarely supported by appropriate study designs and scientific publications. Moreover, it is also widely recognized by practitioners that while standardization offers benefits such as efficiency and quality assurance, it can also pose challenges, including inflexible patient care, "one-size-fits-all" limitations, reduced clinician autonomy, and top-down impositions [3]. Despite growing global interest in these therapies, the development of evidence-based practice guidelines has been limited by the lack of appropriate research designs.

In this context, there is a clear need for methodological approaches that balance consistency with clinical flexibility, providing clear procedural guidance without undermining the individualised nature of Ayurvedic care. Clinical Practice Guidelines (CPGs) offer a promising solution. By integrating available evidence with expert consensus, CPGs provide a structured yet adaptable framework that aligns with Ayurveda's whole-systems approach.

Unlike rigid standardized protocols, CPGs are developed through bottom-up, consensus-driven processes that engage clinicians, researchers, academicians, and, sometimes, patients. They are inherently flexible, and accommodate variations based on geographical, institutional, and patient-specific factors. Moreover, CPGs are not legally binding, allowing practitioners to exercise clinical judgment and deviate from recommendations with appropriate justification. The scope, potential benefits, and challenges of developing CPGs in Ayurveda, along with their political and legal considerations, have been discussed in detail in a previous publication [4].

The Central Council for Research in Ayurvedic Sciences, an apex body for research in Ayurveda under the Government of India, has taken the initiative to develop CPGs for *Panchakarma* and other therapeutic procedures. As an initial step, the procedure of *Marsha Nasya* (MN), a therapy in which medicines mainly having an oil base are instilled in the nose in a comparatively higher dose, was selected, and the study followed a clinical consensus method.

The texts of Ayurveda mention that MN, where the medicine(s) administered through the nasal route, acts on *Sringataka marma* (a vulnerable location(s) at the confluence of the nose, ear, eye, and tongue vessels, which are four in number) and sense organs of the head to alleviate various morbid conditions, especially the head and neck region [5]. Numerous inconsistencies have been identified in the practice of this procedure across its various phases, including patient selection, medication dose calculation, administration method, and follow-up care.

1.1. Need gap analysis

The gap analysis of the current practice of MN is summarised as follows:

1.1.1. Indications for MN

Selecting the patient who will benefit most from MN is essential to the success of the procedure, yet selection criteria have not been investigated thoroughly. The texts of Ayurveda provide an elaborate list of indications and contraindications. However, the current clinical practices require an appropriate interpretation of the classical references. For example, *Nasya* is indicated in *Shira kampa* tremor disorders of the head [6]; it is worth discussing which state of such disorders is an indication and what clinical conditions in the present day may be classified as *Shirakampa*.

1.1.2. Description of MN procedure and follow-up care

Another concern is the lack of systematized methods for practising MN and follow-up care. The diverse interpretations of classical references for the administration of MN have led to considerable variability in clinical practice.

1.1.3. Calculation of the dose of medicine for the MNs

The dose of medicine is calculated in terms of *Bindu,* a basic unit described for the measurement of medicine for MN. *Bindu* refers to standardized volume of medicated liquid employed in Ayurveda nasal drug delivery. It is defined as the amount of medicated liquid (such as oil, herbal juice or decoction) that naturally clings to and drips off the tip of the index finger when the distal and middle phalanges are immersed and then withdrawn from the medicated liquid. In contemporary science, a drop is rounded to precisely 0.05 mL (50 μL, i.e. 20 drops per milliliter) following metric measurements [7,8]. The term *Bindu* is inappropriately interpreted, and the common practice is to equate it to a drop, which is much less than the amount of medicine obtained following the textual method of calculating the *Bindu.*

A scoping review of studies on the effects of *Nasya* revealed that few published studies are available, and these offer limited descriptions of the procedural details [9–42]. A summary of the review is presented in Fig. 1. This review highlights the lack of available published guidelines for performing MN therapy, which forms the background of this study.

1.2. RAND/UCLA Appropriateness Method (RAM)

The RAND/UCLA Appropriateness Method (RAM) is an internationally recognized consensus technique in which a panel of experts codifies appropriate procedures or actions, presented as statements relating to practices and policies. This statistically validated method has been developed to determine proper patient care in situations where strong evidence-based guidelines are not possible [43]. RAM is typically applied in clinical practice, for developing clinical practice guidelines and recommendations.

The present study aims to develop a comprehensive set of evidence-

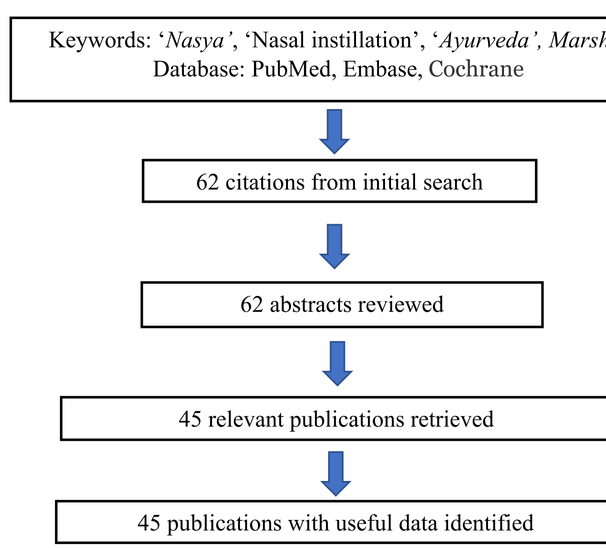

Fig. 1. Summary of scoping rereview.

based, expert-endorsed Best Practice Statements (BPSs) for MN therapy utilizing the RAM.

2. Methods

We used the RAND/UCLA Appropriateness Method (RAM), which combines a systematic summary of available literature related to MN with the collective judgment of experts. This approach required panelists/experts to rate the clarity, appropriateness, and feasibility of statements relating to different components of the therapy using structured rating forms [44]. The study was reviewed and approved by the Institutional Ethics Committee of the study institute (ref. No. F.No.January 3, 2020/NARIP/Tech/2035 dated January 31, 2023). The study was prospectively registered in the Clinical Trial Registry of India (CTRI number: CTRI/2023/05/052422 dated May 09, 2023) and PREPARE (Practice Guideline Registration for Transparency), an international prospective register for clinical practice guidelines. (PREPARE-2024CN612). The methodology is summarized in Fig. 2.

2.1. Clinical consensus panelists

To optimize the possibility of a varied range of viewpoints and skills, we employed purposive sampling across a number of factors to recruit a case-mix sample of panelists. The criteria included distinct geographic locations (i.e., different states,cities, rural and urban areas in India), professional responsibilities and skill sets (i.e., research, clinical, administrative, and academic-based), and expertise across a range of disorders in which MN is administered. The following criteria determined the selection of experts: 1. Registered Ayurveda practitioners who possessed more than ten years of clinical experience practising MN. 2. Those demonstrating a comprehensive understanding of classical Ayurveda texts and published literature related to MN. 13 experts were initially contacted based on the above criteria. But only nine of them agreed to participate, and the same nine participants, (the standard sample size, as recommended by the RAM manual [45]) were present in both the rounds of RAM (Supplementary file 1). They were approached over the telephone for their availability and interest, and when they agreed, the details of the project, including the objectives, were provided through email. Written informed consent was obtained from the prospective panelists before participation. No potential conflicts of interest were identified when the panelists' direct and indirect interests were considered.

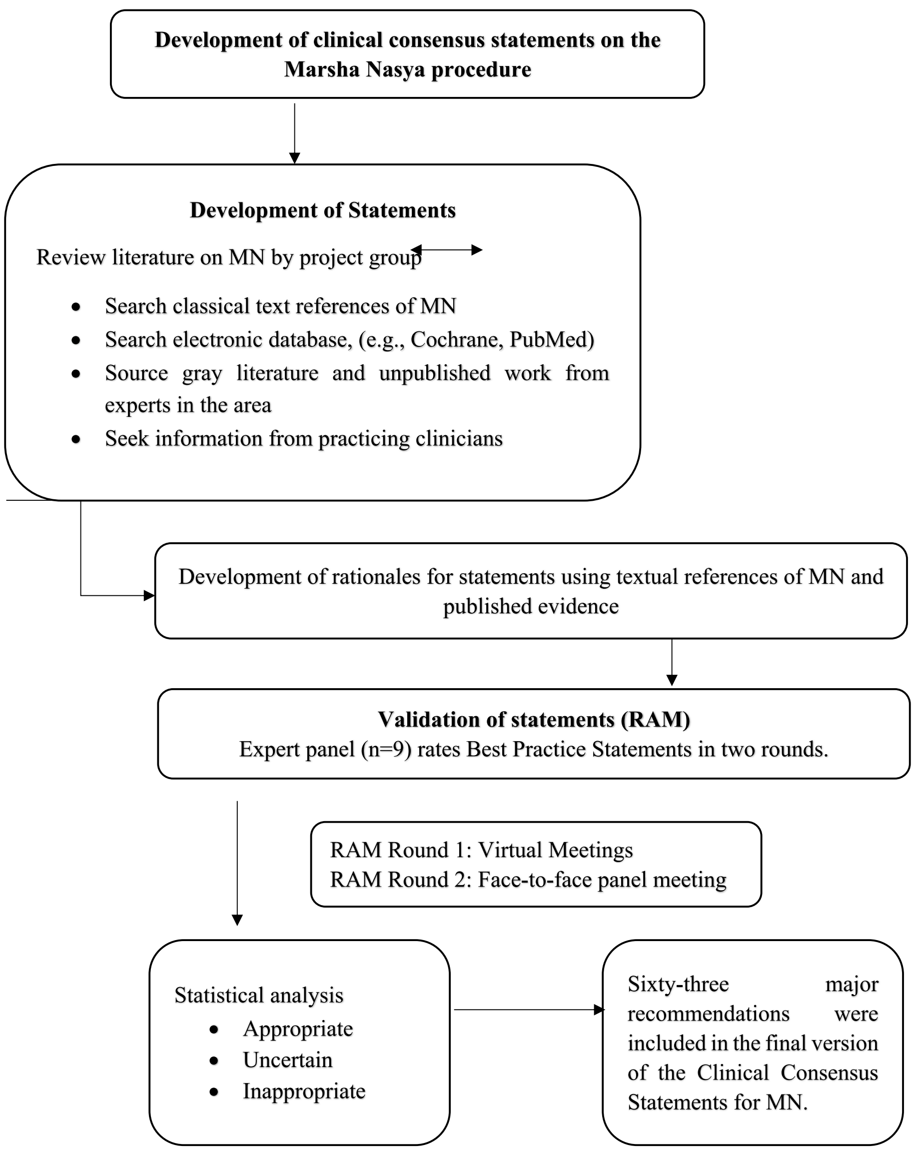

Fig. 2. Flow diagram of the RAND/UCLA appropriateness method (RAM).

2.2. Development of BPSs

The clinical scenarios or best practice statements were formulated based on a literature search of the classical textbooks of Ayurveda, published scientific studies on *Nasya*, inputs from physicians practising *Marsha Nasya* and also a survey study conducted in different prominent healthcare facilities and teaching hospitals of Ayurveda for the practice of MN. Limited published literature was found for evidence synthesis. A total of 75 BPSs were drafted for the first round of RAM rating.

2.3. RAM rating round 1

The first round involved individual ratings of statements which were accomplished through 2–3 hours of virtual meetings. During the virtual meeting, each statement, along with their rationale, was read and discussed. The panelists were also provided with a summary of evidence from published literature and classical texts.

On a scale of one to nine, where nine is the most "appropriate," the panelists rated the "appropriateness" of each statement. They were also requested to provide comments to justify their scores. The comments and scores were entered into a central database file. The panel facilitator verified that all the entries and comments were transferred accurately. The first round of RAM for all panelists was completed over the course of approximately one month.

2.4. Analysis

Recommendations were made based on the panel's degree of agreement (dispersion) and the median rating of appropriateness. Statements classified as "appropriate" had a median panel score in the top tertile (7–9) without disagreement; statements classified as "inappropriate" had a median panel score in the bottom tertile (1–3) without disagreement; and statements classified as "uncertain" had a median panel score between 4 and 6. Agreement was indicated when no more than two panelists rated the statement outside the three-point region (1–3; 4–6; 7–9) containing the median score using the rules for a nine-member panel. When the statement received ratings from at least three panelists in the lower third region (1–3) and at least three panelists in the top third region (7–9), there was disagreement [43]. Panelists were encouraged to offer additional statements or word changes for each statement after the first rating. The investigators performed a content analysis of the remarks to obtain a preliminary understanding of the nature of any concerns the panelists had with the statements. It facilitated the discussion of the statements in the face-to-face round. After the first ranking, the statements that were considered appropriate for the agreement were directly proposed for the second round, possibly with minimal wording or text modifications, as suggested by the panelists. For some statements considered uncertain, a proper bibliography was provided to the panelists, based on which further wording/text modifications were suggested. The statements that were uncertain and not supported by classical literature or had a low level of evidence were eliminated.

2.5. RAM rating round 2

The second round, a one-day face-to-face meeting, was held in Kerala in May 2023, two weeks after the first-round rating of the last panelist was completed. All discussions were audiotaped with the consent of the panelists. Each panelist received a score sheet for this round that included both their initial ratings for each statement and the panel's median score for each statement. The wording of each BPS, along with the rationale, was discussed by the panelists. After the detailed deliberations, the panelists rerated each statement anonymously without further discussion. The analysis of the second-round rating was also performed similarly to that of the RAM rating round 1. For the statements to be deemed "appropriate" and included in the final best practice statement document, they were required to receive a median rating between 7 and 9 from the panelists, with no more than two of them giving them a score below 7. The study was conducted as a partial RAM as only two rounds of rating was conducted to assess the appropriateness of the clinical scenarios. For both the rounds, two of the investigators, one who was expert in *Panchakarma* and the other who had experience in conducting Clinical Consensus studies such as Delphi studies served as the moderators, who supported and guided the experts, and the experts were encouraged to record their independent ratings throughout the process. All panelists were actively engaged in the process. The moderator tried to find a balance between the panelists who dominated and encouraged everyone to participate, and care was taken that the researcher's positionality did not have any influence on the rating of the panelists.

3. Results

A consensus was reached on 63 BPSs across six domains: the general domain, the preparatory procedures domain, the main therapeutic procedure domain, the post-therapy procedures domain, the dosage domain, and the quality assurance domain (Supplementary file 2). All the statements were rated as appropriate and deemed valid for inclusion in the guidelines, except for two statements that discussed the practice of applying *Talam* (a therapeutic procedure involving placing medication over the vertex) and the rubbing of medicated powder over the vertex during *Nasya*. Experts suggested including them as regional practices that require further feasibility studies before being included in clinical practice recommendations that may have national/broader implications.

3.1. Main patterns and arguments

3.1.1. Written consent, performing fitness assessments, and recording daily progress

During the face-to-face discussion, the practices of obtaining written consent, performing a fitness assessment, and meticulously recording the daily progress of the procedure were unanimously agreed upon. The panelists recommended developing fitness and daily assessment forms that incorporate contemporary disease diagnoses, which may render a patient unfit for the procedure.

3.1.2. Time-frequency and dosage form of medicine

The panelists judged that the morning and evening are most suitable for administering MN and approved different frequencies ranging from daily to 'once in two days.' Panelists endorsed the use of oil and ghee as the preferred dosage forms while accepting the use of all four major unctuous substances mentioned in Ayurveda therapeutics for appropriate indications [46].

3.1.3. Dietary restrictions and lifestyle modifications

Panelists explicitly deliberated upon dietary restrictions and lifestyle modifications. The prepared guidelines appropriately addressed the timing of major meals, light food, and the time gap allowed for food after the procedure.

The majority of the panelists reasoned that a head bath should be strictly contraindicated because it may cause imminent *Doshakopa* (vitiation of regulatory functional factors of the body) owing to the involvement of *Shiras* (head), one of the *Trimarma* (major three vital points). However, some panelists argued that in specific regions of the country, it would be extremely difficult to follow the restrictions throughout the course of the treatment, and panelists agreed that the individual contexts of the patient might be emphasised and that the patient may be allowed to use a head bath at specific time intervals if necessary.

3.1.4. Patient preparation, positioning, and other preparatory procedures

Regarding the positioning of the patient, the panelists unanimously judged that administering the medicine in a sitting posture and using a *Nasya* chair was inappropriate, regardless of the patient's disease condition. The panelists argued that *Droni* (a specially designed wooden cot on which patients lay down while performing therapeutic procedures) might not be available in all the treatment facilities; hence, the use of any flat treatment table was endorsed, which can be raised at the leg end. The panelist also suggested that lower-end elevation can be achieved even by raising the patient's legs using a pillow. The panelists agreed that the use of *Jatruurdhwa Abhyanga* (oil application to body parts above the clavicle) should include *Shiro Abhyanga* (oil application to the head) and can also be performed with plain oil such as sesame oil or mustard oil. For sudation, either *Patasweda* (sudation using cloth), *Nadi sweda* (sudation using a pipe-like instrument), or *Hastasweda* (sudation using hands) was recommended as appropriate. *Dhumapana* (medicated smoking), which is not widely practiced as a preparatory procedure, was recommended as appropriate for inclusion in the preparatory procedure domain.

3.1.5. Medicine instillation and post-therapy procedures

All the panelists agreed that the measurement of *Bindu* should not be equated with a drop. Even though *Gokarna* (the small vessel used to drop medicine to the nostril) was mentioned in the texts, the panelists agreed that *Gokarna, Pichu* (tampon), spoon, or syringe could also be used to instill the medicine. When some of the panelists expressed their reservations about using the spoon or syringe, the panel judged that a continuous flow of medicine should be ensured while administering the medicine irrespective of the instrument used, and emphasized that it should not be administered as drops. Regarding the position of the therapist/physician who administers the medicine, standing behind the patient's head end is considered appropriate, contrary to the practice where the therapist/physician stands at the side of the patient. Contrary to the common practice of single instillation, multiple instillations are recommended. The patient is advised to remain in the supine position for approximately one minute forty seconds, and gargling with warm water is recommended following the inhalation of medicated smoke.

3.1.6. Dose calculations and quality indicators

The panelists agreed that the dose should be calculated as follows: the patient is asked to dip the index finger in unctuous medicine up to the proximal interphalangeal joint, and the amount that drips out of the finger is to be measured. This should be repeated three times. The average amount of medicine should be calculated as one *Bindu* specific to that particular unctuous medicine. The dosage schedule can be followed as either a constant fixed dose or an increasing dose throughout the course of treatment.

All the panelists strongly agreed that the procedure should only be administered under the supervision of a qualified Ayurveda physician, and all the therapists and physicians involved in administering the procedure should be trained in basic life support (BLS). The strict implementation of sterilization and sanitization measures has also been strongly endorsed. The external peer review of the document has also endorsed the recommendations in the guidelines.

4. Discussion

With the advent of modern pedagogical approaches and recent developments such as quality accreditation and insurance coverage in Ayurveda, there is an increasing need to establish clear and practical guidelines for the therapeutic procedures—particularly the *Panchakarma* therapies. Although efforts have been made in the past to standardize these procedures, such initiatives often encounter inherent challenges. Notably, standardization can conflict with the foundational principles of Ayurveda practice, which emphasize the physician's autonomy and adaptability within the classical framework. Further, it also overlooks the regional variations, which is one of the inherent strengths of Traditional Medicine systems such as Ayurveda.

The tradition of discussion and consensus is deeply rooted in Ayurveda classics [47] which encouraged debate as a means to arrive at clinical conclusions and therefore, the RAM methodology was considered as an alternative in this direction. As an initial exploration, this study presents the consensus statements which have evolved as a result of such an exercise. Further, it helps the practitioners to know the learned consensus opinion of experts on the various aspects relating to the procedure, thus bringing better clarity in clinical practice. In contrast to standardization, the process of development of Clinical Practice Guidelines has a broader purpose for clinical practice, research and education domains the summary of which is presented in Fig. 3 [4].

Expert consensus is vital for guiding decision-making where empirical evidence is scarce or inconclusive. Among the four traditional consensus methods - Delphi, Nominal Group, RAND/UCLA Appropriateness Method, and Consensus Conference - the Delphi and RAM methods are the most commonly utilized. The RAM method was chosen for this study as it focuses on clinical scenarios, making it particularly ideal for treatment indications. In contrast, the Delphi method is based on question-and-answer sets, which are ideal for addressing issues related to terminology, diagnosis, planning, and strategy [48].

In Ayurveda, published CPGs that follow a methodical research approach are rare [49]. To the best of our knowledge, this is the first-ever attempt to publish Clinical Consensus Statements/Clinical Practice Guidelines for any therapeutic procedure in Ayurveda, which has followed a systematic development process. While a general set of guidelines for safer *Panchakarma* practices in non-COVID-19 clinical care was published during COVID-19, they lacked a specific study design [2]. Clinical guidelines in Western medicine have been developed to reflect the current evidence base. However, in Ayurveda, the scope of developing clinical practice guidelines limits itself to codifying the existing clinical practices in a specific area in light of the opinions of experts and the supporting literature available in classical texts.

Owing to the fact that the level of published evidence (published scientific studies) is lower, the panelists recommended that it be titled as a clinical consensus statements rather than Clinical Practice Guideline, which would need further feasibility studies under different clinical settings [50]. Many of the BPSs in the guidelines can serve as potential leads for further, more extensive clinical studies.

As the daily practice of MN is even more complex than that of theoretical scenarios, the BPSs was simplified and made suitable for an average clinical facility where MN can be administered. The panelists underscored that individual patient characteristics could modify the suitability of these statements, perhaps resulting in a clinical conclusion that deviates from the panel's assessment. End users may choose to utilize statements in different ways. Clinicians making decisions about MN may focus on the relevant recommendations, whereas hospital administrators or policymakers may examine the supporting evidence to inform broader decision-making.

4.1. Limitations of the study

A few limitations should be considered when assessing the outcomes of this study.

One of the major limitations identified is that the Grading of Recommendations Assessment, Development, and Evaluation (GRADE) system, which is commonly used to assess the quality of evidence and the strength of the recommendations, could not be applied because the step of the "Evidence to Decision" (EtD) framework was not adopted. The GRADE approach to rating the quality of evidence begins with the study design, and ancient textbooks and expert opinions are not considered a quality of evidence category [51,52].

The RIGHT statements, which are the standard reporting guidelines for reporting clinical practice guidelines, could not be fully applied because they do not acknowledge the textual evidence and experience of

- Enable judicious accommodation of practice variations with appropriate clinical reasoning—for Nasya and other procedures.
- Recommendations are framed around an average patient, clinician, and facility, supporting cost-effective care.
- Enhance clarity on recommended actions, promoting optimal use of resources.
- Support the adoption of best practices, facilitating quality control in therapeutic procedures.
- Aid in achieving quality certifications from bodies like NABH and NABL under the Quality Council of India (QCI).
- Help curb misleading advertisements related to therapies like Nasya, thereby improving credibility and public trust in Ayurveda.

Fig. 3. Additional benefits the Clinical Consensus Statements (CCSs) offer to the field of Ayurveda.

expert physicians [53]. Supplementary file 3 shows an attempt to report this study using the RIGHT checklist. We have found that it is challenging to reflect the principles/major characteristics of traditional Indian medicine when RIGHT statements are used to report the development of guidelines. This underscores the importance of adapting and broadening existing reporting guidelines within conventional medicine to accommodate the requirements of Ayurveda, similar to the RIGHT Extension Statement for Traditional Chinese Medicine (TCM) and AGREE II for TCM which demonstrate the inclusion of characteristics unique to TCM CPGs [54,55].

Further, since MN involves instillation in a comparatively higher dose, these guidelines are not applicable to other variants of *Nasya* or to other similar procedures such as the instillation of nasal drops.

4.2. Implications of the results and future research

Feasibility studies of the developed guidelines are already initiated in three distinct geographic clinical settings in India, and the results will be published in a separate study. As stated earlier, many BPSs included in clinical consensus statements can serve as potential leads for future research. Observational studies such as prospective cohort studies and Randomised Controlled Trials (RCTs), as a part of Comparative Effectiveness Research (CER), may be designed to explore the efficacy and feasibility of recommended practice guidelines. The outcomes of CER may be used to revise and update the practice guidelines. It can contribute to the quality standards prepared for healthcare facilities in Ayurveda by national accreditation bodies such as the National Accreditation Board for Hospitals & Healthcare Providers (NABH). These can act as a reference point in auditing healthcare facilities to rate them against best practices.

As this study represents one of the initial efforts toward developing Clinical Practice Guidelines (CPGs) in Ayurveda, the authors urge readers to interpret the study's findings in light of the following considerations. Fig. 4.

4.3. Plans for dissemination

The dissemination plans include making these statements available on the websites of major stakeholder institutions such as the Ministry of Ayush (MoA), the Central Council for Research in Ayurvedic Sciences (CCRAS), and the National Commission for Indian Systems of Medicine (NCISM), reaching out to academic institutes, and other healthcare

- CPGs in bio-medicine rely on high-quality published research, whereas Ayurveda often draws from expert opinion, and classical texts and real world (clinical) data (RWD). As current evidence grading systems like GRADE classify such sources as low-quality, the recommendations may require further feasibility testing in diverse clinical settings.
- Ideally, consensus panels should include multidisciplinary stakeholders. However, due to the early stage of CPG development in Ayurveda, the panel was limited to a small number of to clinicians, academicians, and researchers.
- Only nine experts from various parts of India participated. Certain regions—such as the far North and Northeast India—were underrepresented, potentially overlooking regional variations in practice that may be present.
- The Clinical Consensus Statements (CCSs) are not legally binding and fully allows physician's clinical judgment. Physicians may deviate from them based on clinical judgment, provided such deviations are well justified.

Fig. 4. Considerations for readers while interpreting the study's findings.

facilities for its implementation. Initiating training programs and incorporating them into the curriculum are some of the other potential dissemination measures.

4.4. Guideline review and anticipated update

The update for these recommendations is anticipated to occur in 3 years or earlier if practice-changing evidence becomes available.

5. Conclusion

The final 63 recommendations represent the important procedural aspects of MN. The detailed, expert endorsed and evidence based statements can be the basis for further clinical validation and practice of MN. The study may serve as a model for developing evidence-informed clinical practice guidelines for *Panchakarma* therapies and other therapeutic procedures in Ayurveda and can offer valuable guidance to clinicians, researchers, students, and policymakers.

Ethics approval and consent to participate

The study was reviewed and approved by the Institutional Ethics Committee of the study institute (ref. No. F.No.January 3, 2020/NARIP/Tech/2035 dated January 31, 2023).

Author contributions

DN: Conceptualization, Project administration, Writing – review & editing. **AK:** Conceptualization, Methodology, Investigation, Formal Analysis. Writing – original draft. **PK:** Investigation. **SL:** Investigation. **SK:** Methodology, Writing – review & editing. Visualization. **BC:** Conceptualization, Writing – review & editing. **NS:** Conceptualization, Supervision, Writing – review & editing. **RN:** Conceptualization, Supervision. All authors read, provided feedback, and approved the final manuscript.

Declaration of generative AI in scientific writing

None.

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Conflict of interest

The authors declare that they have no known competing interests.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.jaim.2025.101298.

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